Rejection Sampling: Gamma(2,1) Target with Exponential(0.8) Proposal

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May 12, 2026

Problem Statement

We aim to perform rejection sampling to draw samples from a target distribution $f(x)$ using an instrumental (proposal) distribution $g(x)$.

- **Target:** $f(x) = \text{Gamma}(2,1) = \frac{xe^{-x}}{6e^{-5}}$ (normalized over the relevant interval)
- **Proposal:** $g(x) = \text{Exp}(0.8) = 0.8e^{-0.8x}$

1. Finding the Ratio of Target to Proposal

To find the scaling factor M such that $f(x) \leq Mg(x)$ for all x, we first calculate the ratio:

$$\begin{aligned}
\frac{f(x)}{g(x)} &= \frac{\frac{xe^{-x}}{6e^{-5}}}{\frac{4}{5}e^{-\frac{4}{5}x}} \\
&= \frac{1}{6e^{-5}}\left(\frac{xe^{-x}}{\frac{4}{5}e^{-\frac{4}{5}x}}\right) \\
&= \frac{5}{24e^{-5}}(x)e^{-x-(-\frac{4}{5}x)} \\
&= \frac{5}{24e^{-5}}(x)e^{-\frac{1}{5}x}
\end{aligned}$$

Thus, our ratio function is:

$$\frac{f(x)}{g(x)} = \frac{5xe^{-\frac{1}{5}x}}{24e^{-5}}$$

2. Maximizing the Ratio

We find the value of x that maximizes this ratio by taking the derivative with respect to x and setting it to zero. Using the product rule with $u = x$ and $v = e^{-\frac{1}{5}x}$:

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{5}{24e^{-5}}\left[\left(-\frac{1}{5}\right)xe^{-\frac{1}{5}x} + e^{-\frac{1}{5}x}\right]$$

Setting the derivative to zero:

$$0 = \frac{5}{24e^{-5}}e^{-\frac{1}{5}x}\left(-\frac{1}{5}x + 1\right)$$

Since $\frac{5}{24e^{-5}}e^{-\frac{1}{5}x} \neq 0$ for all finite x, we solve:

$$\begin{aligned}
-\frac{1}{5}x + 1 &= 0 \\
-\frac{1}{5}x &= -1 \\
x &= 5
\end{aligned}$$

3. Calculating the Scaling Factor M

The maximum ratio occurs at $x = 5$. We calculate M by evaluating the ratio at this point:

$$\begin{aligned}
M &= \frac{f(5)}{g(5)} \\
&= \frac{\frac{5e^{-5}}{6e^{-5}}}{\frac{4}{5}e^{-\frac{4}{5}(5)}} \\
&= \frac{5/6}{4/5e^{-4}} \\
M &\approx 56.873
\end{aligned}$$